

8.4.1 LONGITUDINAL ACOUSTIC PHONONS AND THE DEFORMATION POTENTIAL

Longitudinal acoustic phonons are like sound waves for long wavelengths (small q). The simplest coupling to electrons for such phonons is through the deformation potential. The phonon compresses and dilates alternating regions of the solid. A uniform compression or dilation of the crystal causes the edge of each electronic energy band to move up or down proportionally to the strain. The constant of proportionality is called the *deformation potential* Ξ . (A more complicated expression is needed if the bands are degenerate or lack spherical symmetry.) Longitudinal strain is defined as the fractional increase in length of the object, so for one unit cell of the chain (Figure 2.21(a)) it is given by $(u_j - u_{j-1})/a$. For long wavelengths we can treat the one-dimensional chain of atoms as a continuum and the strain becomes a derivative. The displacement of an atom at z is given by equation (2.30), $u(z) = U_0 \cos(qz - \omega_q t)$, so the longitudinal strain is

$$e(z) = \frac{\partial u}{\partial z} = -U_0 q \sin(qz - \omega_q t). \quad (8.45)$$

The potential energy can be calculated as though the strain were uniform in each region provided that the wavelength is long. This gives

$$V(z) = \Xi e(z) = -U_0 q \Xi \sin(qz - \omega_q t) = -\sqrt{\frac{2\hbar}{\Omega \rho \omega_q}} q \Xi \sin(qz - \omega_q t), \quad (8.46)$$

where the amplitude U_0 for a single phonon has been inserted from equation (2.30). Since $\omega_q = v_s q$ for long-wavelength acoustic phonons, this finally becomes

$$\hat{V}(z, t) = i \sqrt{\frac{\hbar q}{2\Omega \rho v_s}} \Xi (e^{iqz} e^{-i\omega_q t} - e^{-iqz} e^{i\omega_q t}). \quad (8.47)$$

This is the form of the perturbing potential caused by the phonon to be used in Fermi's golden rule.

One more ingredient, the number of phonons, is needed to get the total scattering rates. The number of phonons that occupy a mode of wave vector $\mathbf{Q}$ at equilibrium is given by the Bose–Einstein distribution (1.127):

$$N_{\mathbf{Q}} = \left[\exp\left(\frac{\hbar \omega_{\mathbf{Q}}}{k_B T}\right) - 1 \right]^{-1}. \quad (8.48)$$

The rate for a single phonon must be multiplied by the number of phonons in a mode $N_{\mathbf{Q}}$ to calculate the total rate of scattering due to absorption. The factor for emission of phonons is $N_{\mathbf{Q}} + 1$; the 1 describes *spontaneous* emission while the $N_{\mathbf{Q}}$ describes *stimulated* emission. (Spontaneous emission can instead be viewed

as emission stimulated by vacuum fluctuations.) This asymmetry between the rates for absorption and emission is vital to ensure that electrons and phonons remain in equilibrium with one another.

Scattering in low-dimensional systems is complicated because the phonons usually retain their three-dimensional nature, even if the electrons are confined. We shall therefore look first at the interaction between three-dimensional electrons and phonons and defer the two-dimensional electron gas to Section 9.6.3. Put the electrons in a finite box of volume Ω where the wave functions are plane waves, scattering from $\mathbf{K}$ to $\mathbf{K}'$. Start with the first term in equation (8.47). This depends on time like $\exp(-i\omega_Q t)$ and causes the electron to absorb energy from the phonon. The condition for energy conservation in the δ -function is therefore $\varepsilon(\mathbf{K}') = \varepsilon(\mathbf{K}) + \hbar\omega_Q$. The matrix element is

$$V_{\mathbf{K}'\mathbf{K}}^+ = i\sqrt{\frac{\hbar Q}{2\Omega\rho v_s}} \Xi \frac{1}{\Omega} \int_{\Omega} e^{-i\mathbf{K}'\cdot\mathbf{R}} e^{i\mathbf{Q}\cdot\mathbf{R}} e^{i\mathbf{K}\cdot\mathbf{R}} d^3\mathbf{R}. \quad (8.49)$$

The integral gives Ω , which cancels the $1/\Omega$ from the normalization of the wave functions, if the wave vectors sum to $\mathbf{0}$, and zero otherwise. This requires $\mathbf{K}' = \mathbf{K} + \mathbf{Q}$, the expected result for the conservation of momentum. We must also include a factor N_Q to account for the number of phonons for absorption. Thus the scattering rate from $\mathbf{K}$ to $\mathbf{K}'$ caused by the absorption of longitudinal acoustic phonons is

$$W_{\mathbf{K}'\mathbf{K}}^+ = \frac{2\pi}{\hbar} N_Q \frac{\hbar Q}{2\Omega\rho v_s} \Xi^2 \delta_{\mathbf{K}',\mathbf{K}+\mathbf{Q}} \delta[\varepsilon(\mathbf{K} + \mathbf{Q}) - \varepsilon(\mathbf{K}) - \hbar\omega_Q]. \quad (8.50)$$

The relation $\mathbf{K}' = \mathbf{K} + \mathbf{Q}$ enforced by the δ -function for the wave vectors has been used to replace $\mathbf{K}'$ inside the δ -function for energy. The electron has gained wave vector $\mathbf{Q}$ and energy $\hbar\omega_Q = \hbar v_s q$ from the phonon.

The second half of the perturbation (equation 8.47) causes the electron to lose both momentum and energy to the phonon. The only differences are the signs in the δ -functions and the $N_Q + 1$ in the phonon occupation factor to include spontaneous as well as stimulated emission. Thus

$$W_{\mathbf{K}'\mathbf{K}}^- = \frac{2\pi}{\hbar} (N_Q + 1) \frac{\hbar Q}{2\Omega\rho v_s} \Xi^2 \delta_{\mathbf{K}',\mathbf{K}-\mathbf{Q}} \delta[\varepsilon(\mathbf{K} - \mathbf{Q}) - \varepsilon(\mathbf{K}) + \hbar\omega_Q]. \quad (8.51)$$

Unfortunately the change in energy prevents us from writing a simple formula for the contribution of phonons to the transport scattering rate, as we did for impurities. However, we can find out whether acoustic phonons carry away significant energy. Consider a 2DEG of density $n_{2D} = 3 \times 10^{15} \text{ m}^{-2}$, for which $E_F \approx 10 \text{ meV}$ and $k_F \approx 0.14 \text{ nm}^{-1}$. Backscattering provides the phonon with the largest wave vector, $q = 2k_F$, whose energy is $\hbar\omega_{2k_F}$. The velocity of sound is around 5 km s^{-1} , so the energy is $2\hbar v_s k_F \approx 0.9 \text{ meV}$. The conclusion is that acoustic phonons do not carry

away much energy, and they are often treated in the *quasi-elastic approximation* where this energy is neglected. The same conclusion is reached for non-degenerate electrons at room temperature.

As a simple example we can calculate the total scattering rate for an electron in a three-dimensional gas using the quasi-elastic approximation. The condition for energy conservation becomes $\varepsilon(\mathbf{K} + \mathbf{Q}) = \varepsilon(\mathbf{K})$, as for impurities. Also, the number of phonons in the relevant modes is large, so we can use the non-degenerate limit of the Bose distribution, $N_{\mathbf{Q}} \approx (N_{\mathbf{Q}} + 1) \approx k_B T / \hbar \omega_{\mathbf{Q}} \gg 1$. These approximations mean that the rates for emission and absorption are nearly equal and together give the total rate for scattering from $\mathbf{K}$ to $\mathbf{K} + \mathbf{Q}$,

$$\begin{aligned} W_{\mathbf{K}+\mathbf{Q},\mathbf{K}} &= 2 \frac{2\pi}{\hbar} \frac{k_B T}{\hbar v_s Q} \frac{\hbar Q}{2\Omega \rho v_s} \Xi^2 \delta[\varepsilon(\mathbf{K} + \mathbf{Q}) - \varepsilon(\mathbf{K})] \\ &= \frac{2\pi}{\hbar} \frac{1}{\Omega} \frac{\Xi^2 k_B T}{\rho v_s^2} \delta[\varepsilon(\mathbf{K} + \mathbf{Q}) - \varepsilon(\mathbf{K})]. \end{aligned} \quad (8.52)$$

After all the simplifications this resembles the scattering rate (8.23) for impurities. In fact this is the particularly simple case of isotropic scattering because $\mathbf{Q}$ has vanished from the matrix element. Summing over all $\mathbf{Q}$ gives the total scattering rate for an electron with wave vector $\mathbf{K}$,

$$\frac{1}{\tau} = \frac{2\pi}{\hbar} \frac{\Xi^2 k_B T}{\rho v_s^2} \left\{ \frac{1}{\Omega} \sum_{\mathbf{Q}} \delta[\varepsilon(\mathbf{K} + \mathbf{Q}) - \varepsilon(\mathbf{K})] \right\}. \quad (8.53)$$

The sum gives the density of states as in equation (1.97) except that there is no summation over spin. We could have reached this result more directly by using the 'density-of-states' form of Fermi's golden rule. Thus the scattering rate reduces to

$$\frac{1}{\tau} = \frac{\pi \Xi^2 k_B T n[\varepsilon(\mathbf{K})]}{\hbar \rho v_s^2}. \quad (8.54)$$

The single-particle and transport lifetimes are identical for isotropic scattering so no subscript is needed on τ . A typical energy in a non-degenerate three-dimensional electron gas is $\varepsilon(\mathbf{k}) = \frac{3}{2} k_B T$. The density of states $n(E) \propto \sqrt{E} \propto \sqrt{T}$, so we reach the well-known result $\mu \propto T^{-3/2}$.

There are other mechanisms by which electrons and LA phonons interact. For example, the motion of ionic cores generates a magnetic field. More important, strain may generate a piezoelectric potential in compound semiconductors, and there remains considerable dispute over the relative importance of deformation-potential and piezoelectric coupling in GaAs.

8.4.2 LONGITUDINAL OPTIC PHONONS AND POLAR COUPLING

Optic phonons occur in crystals with more than one atom per unit cell (Section 2.8.3). They provide a contrast to acoustic phonons because their energy is large (36 meV at $Q = 0$ in GaAs). They also illustrate a quite different way of coupling to the electrons. The name 'optic' arises because the two atoms in a unit cell move in opposite directions (Figure 2.24(b)). This sets up an electric field if the atoms carry a charge, and we shall consider only this polar case, which includes the III-V semiconductors.

Since the *relative* displacement of the two atoms in a unit cell is the relevant coordinate for an optic phonon, denote this by $u_j(t) = U_0 \cos(\mathbf{Q} \cdot \mathbf{R} - \omega_Q t)$. The dispersion relation (Figure 2.23) is nearly flat around $\mathbf{Q} = 0$ so we can put $\omega_Q = \omega_{LO}$, a constant. The prefactor U_0 for a single phonon can be found in the same way as for the acoustic phonon in Section 2.8.1, with the result

$$U_0 = \sqrt{\frac{2\hbar}{N_{\text{cells}}\mu\omega_{LO}}} = \sqrt{\frac{2\hbar}{\Omega n_{\text{cells}}\mu\omega_{LO}}}, \quad (8.55)$$

where n_{cells} is the number of unit cells (pairs of ions) per unit volume. This is similar to the result (2.29) for acoustic phonons, the main difference being the appearance of the *reduced mass* defined by

$$\frac{1}{\mu} = \frac{1}{m} + \frac{1}{M}, \quad (8.56)$$

where m and M are the two individual masses. The two ionic masses are very similar in GaAs and $\mu \approx \frac{1}{2}m_{\text{ion}}$.

Let the two ions carry opposite charges $\pm Q_{\text{eff}}e$; materials such as GaAs are only weakly ionic so the effective charge $Q_{\text{eff}} \ll 1$. As each pair of ions vibrates, it sets up an electric dipole $p(t) = Q_{\text{eff}}eu(t)$. This in turn generates a polarization field $P(t) = n_{\text{cells}}p(t)$. No free charge has been introduced so the displacement $\mathbf{D} = 0$ and the general relation $\mathbf{D} = \epsilon_0\mathbf{E} + \mathbf{P}$ reduces to $\mathbf{E} = -\mathbf{P}/\epsilon_0$. The electric potential is the integral of E , which introduces a factor of $1/Q$. The potential energy for an electron is finally given by

$$V(\mathbf{R}, t) = \frac{e^2 Q_{\text{eff}}}{\epsilon_0 Q} \sqrt{\frac{2\hbar n_{\text{cells}}}{\Omega \mu \omega_{LO}}} \sin(\mathbf{Q} \cdot \mathbf{R} - \omega_{LO} t). \quad (8.57)$$

The scattering rate can then be calculated as for acoustic phonons, and the rate for absorption of optic phonons is

$$W_{\mathbf{K}'\mathbf{K}}^+ = \frac{2\pi}{\hbar} N_{LO} \left(\frac{e^2}{\epsilon_0}\right)^2 \frac{\hbar n_{\text{cells}} Q_{\text{eff}}^2}{2\Omega \mu \omega_{LO} Q^2} \delta_{\mathbf{K}', \mathbf{K}+\mathbf{Q}} \delta[\epsilon(\mathbf{K} + \mathbf{Q}) - \epsilon(\mathbf{K}) - \hbar\omega_{LO}], \quad (8.58)$$

where N_{LO} is the occupation number of LO phonons given by the Bose–Einstein distribution.

There are two important differences from the result for acoustic phonons. First is the factor of $1/Q^2$. This causes the scattering rate of polar optic phonons to fall off rapidly as a function of Q , in contrast with acoustic phonons, where the coupling grows like Q . Second, the energy of $\hbar\omega_{LO}$ exchanged with the phonon is large. This means that absorption of optic phonons is significant only at high temperatures (typically above 77 K), as they are frozen out below this. However, spontaneous emission is possible at lower temperatures if the electron has sufficient energy, and optic phonons cool hot electrons effectively. The scattering rate as a function of the energy of an electron increases rapidly at the threshold $\varepsilon(K) = \hbar\omega_{LO}$ where spontaneous emission becomes possible, but then declines slowly as the factor of $1/Q^2$ causes the coupling to become weaker. Another important effect of LO phonons is to ionize excitons at room temperature, which we shall consider in Section 10.7.

Although the effective charge Q_{eff} seems an obvious factor to include in the scattering rate, it is usually expressed in terms of the dielectric constant at low and high frequencies:

$$Q_{\text{eff}}^2 = \frac{\epsilon_0 \omega_{LO}^2 \mu}{e^2 n_{\text{cells}}} \left[\frac{1}{\epsilon(\infty)} - \frac{1}{\epsilon(0)} \right]. \quad (8.59)$$

Only the electrons can respond to an electric field at frequencies much higher than ω_{LO} because the ions are too heavy, giving a dielectric constant $\epsilon(\infty)$. At frequencies below ω_{LO} the ions can also respond to an applied field and, being charged, make an additional contribution to the screening, so the dielectric constant $\epsilon(0)$ becomes larger. For GaAs $\epsilon(0) = 12.90$ and $\epsilon(\infty) = 10.92$. The factor in front of the square brackets in equation (8.59) is close to 3 for the III–V semiconductors, so we find $Q_{\text{eff}} \approx 0.2$ in GaAs. The same dielectric constants also appear in the *Lyddane–Sachs–Teller* relation between the frequencies of transverse and longitudinal optic phonons,

$$\frac{\omega_{TO}^2}{\omega_{LO}^2} = \frac{\epsilon(\infty)}{\epsilon(0)}. \quad (8.60)$$

The frequencies are different because the electric field set up by longitudinal vibrations gives an additional restoring force that raises their frequency. Purely transverse vibrations do not generate dipoles so this effect is absent.

As in the case of acoustic phonons, there are other mechanisms by which optic phonons can interact with electrons. A deformation potential can again be defined and is important in elemental semiconductors where there is no polar coupling.

Before leaving phonon scattering we should take a closer look at the matrix elements because our treatment of the wave functions has been rather sloppy. We included only a plane wave, in the spirit of effective-mass theory, but omitted the Bloch function $u_c(\mathbf{R})$ that should multiply it. The integral in equation (8.49)

should be replaced by

$$\begin{aligned} & \int_{\Omega} u_c^*(\mathbf{R}) e^{-i\mathbf{K}' \cdot \mathbf{R}} e^{i\mathbf{Q} \cdot \mathbf{R}} u_c(\mathbf{R}) e^{i\mathbf{K} \cdot \mathbf{R}} d^3\mathbf{R} \\ &= \sum_{mn} u_{cm}^* u_{cn} \int_{\Omega} e^{-i(\mathbf{K}' + \mathbf{G}_m) \cdot \mathbf{R}} e^{i\mathbf{Q} \cdot \mathbf{R}} e^{i(\mathbf{K} + \mathbf{G}_n) \cdot \mathbf{R}} d^3\mathbf{R}. \end{aligned} \quad (8.61)$$

The Bloch functions have the periodicity of the lattice and have therefore been expanded on the right-hand side as Fourier series with the wave vectors $\mathbf{G}_n$ of the reciprocal lattice. The general term survives the integration if $\mathbf{K}' = \mathbf{K} + \mathbf{Q} + \mathbf{G}_n - \mathbf{G}_m$. Thus $\mathbf{K}'$ and $\mathbf{K} + \mathbf{Q}$ need not be identical but may differ by a reciprocal lattice vector. This is called *umklapp scattering*. It is particularly important for intervalley scattering in Si, where the shortest route between two *X*-valleys may involve a reciprocal lattice vector. Umklapp scattering is shown naturally in the repeated zone scheme of band structure.

A full treatment of electron-phonon scattering has to go far beyond this brief discussion. The interaction is screened by electrons or holes if there is a dense gas of them, and this can have a marked effect on the coupling. The scattering of holes is difficult to calculate because of the complicated nature of the valence bands. Intervalley scattering must also be included in a material with degenerate valleys in the conduction band such as Si, and this may be umklapp scattering as just discussed. Scattering from Γ to higher valleys (*X* and *L*) also becomes important for hot electrons in GaAs, where it gives rise to the Gunn effect (Section 2.6.4.3).

8.6 Optical Absorption

The optical properties of low-dimensional systems provide their most widespread applications and fill Chapter 10. Here we shall use the golden rule to derive a general formula for optical absorption and apply it to two contrasting examples: transitions from the valence band to the conduction band where there is a continuous range of absorption, and absorption between the levels of a quantum well where only discrete frequencies are permitted. We face the same problem as in electron-phonon scattering, which is that the photons should be quantized as well as the electrons. Here we shall ignore quantization of the photon field and treat it simply as a classical electromagnetic wave. This is adequate for traditional measurements of optical absorption but would need to be improved for lasers where a correct treatment of spontaneous and stimulated emission is needed. Also, the calculations are limited to linear effects and do not describe phenomena such as frequency doubling seen with light of very high intensity.

For clarity, the formulas will be written down as though they describe an infinite homogeneous medium. This is obviously not true for a heterostructure and one should calculate the electromagnetic fields appropriately, applying matching

conditions at interfaces. We shall assume that this has been done, using the laws of classical electromagnetism. In practice this step may be far from trivial, even for common structures such as a rib wave guide.

8.5.1 MACROSCOPIC EQUATIONS

In classical electromagnetism, the response to an electric field is described by both a current $\mathbf{J}$ and a polarization $\mathbf{P}$, which is usually absorbed into the displacement $\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$. Let the electric field be a single Fourier component with angular frequency ω and wave vector $\mathbf{Q}$, given by the real part of $\mathbf{E}(\mathbf{R}, t) = \mathbf{E}(\mathbf{Q}, \omega) \exp[i(\mathbf{Q} \cdot \mathbf{R} - \omega t)]$. A reminder about signs is in order here. This book employs the 'quantum-mechanics convention' for the dependence on time, $e^{-i\omega t}$, but standard notation in electrical engineering is $e^{j\omega t}$. Unfortunately this difference in sign is pervasive. Users of $e^{j\omega t}$ often define $\tilde{\epsilon}_r = \epsilon_1 - j\epsilon_2$ to ensure that the physically meaningful sign of ϵ_2 , which represents dissipation, is unchanged. A general rule is to change j to $-j$ everywhere. Take care!

The fourth Maxwell equation is

$$\text{curl } \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t} + \mathbf{J} = -i\omega \mathbf{D} + \mathbf{J}, \quad (8.62)$$

where the second form follows from the harmonic dependence on time. The fields $\mathbf{D}$ and $\mathbf{J}$ are given by the constitutive relations $\mathbf{D} = \epsilon_0 \epsilon_r \mathbf{E}$ and $\mathbf{J} = \sigma \mathbf{E}$, where ϵ_r is the dielectric function (relative permittivity) and σ is the conductivity. Both are functions of frequency and wave vector; they should be second-rank tensors rather than scalars but we shall assume that the material is isotropic so that this complication can be neglected for now. Although this simplification holds for the cubic semiconductors, it is not true for heterostructures such as quantum wells where we shall see that the response to an electric field normal to the well may be quite different from the response to a parallel field.

It is inconvenient to have two quantities so they are usually combined into a complex dielectric function $\tilde{\epsilon}_r$ or a complex conductivity $\tilde{\sigma}$. Then the fourth Maxwell equation can be written in the two forms

$$\text{curl } \mathbf{H} = (-i\omega\epsilon_0\epsilon_r + \sigma)\mathbf{E} = -i\omega\epsilon_0\tilde{\epsilon}_r\mathbf{E} = (\tilde{\sigma} - i\omega\epsilon_0)\mathbf{E}. \quad (8.63)$$

The form with conductivity has two terms because free space has $\tilde{\epsilon}_r = 1$ but $\tilde{\sigma} = 0$. The complex dielectric function and conductivity are related by

$$\tilde{\sigma} = -i\omega\epsilon_0(\tilde{\epsilon}_r - 1), \quad \tilde{\epsilon}_r = 1 + \frac{i}{\omega\epsilon_0}\tilde{\sigma}. \quad (8.64)$$

They are entirely equivalent, and one uses whichever seems the more appropriate. Their real and imaginary parts are denoted $\tilde{\epsilon}_r = \epsilon_1 + i\epsilon_2$ and $\tilde{\sigma} = \sigma_1 + i\sigma_2$. The

dielectric function and conductivity that we introduced first are just the real parts of the corresponding complex quantities.

There is an important physical distinction between the real and imaginary parts: σ_1 and ϵ_2 describe conduction, which is a dissipative process, whereas the other part describes polarization, which does not absorb energy. This means that we usually expect $\sigma_1 > 0$, as a negative sign would imply that energy was released when the system was stimulated. Another way of looking at this is to note that σ_1 gives a current in phase with the electric field, and hence dissipation, whereas σ_2 gives a current in quadrature, which is associated with an oscillating transfer of energy but no flow on average.

The complex refractive index $\tilde{n}_r = n_r + i\kappa_r$ is often used for optical properties, defined by $\tilde{\epsilon}_r = \tilde{n}_r^2$. It relates the wave number and frequency of a light wave through $k = \tilde{n}_r \omega / c = (n_r + i\kappa_r) \omega / c$. Thus n_r affects the wavelength and velocity while κ_r gives the rate at which the wave decays with distance. The real and imaginary parts are related by $\epsilon_1 = n_r^2 - \kappa_r^2$ and $\epsilon_2 = 2n_r\kappa_r$. Rapid attenuation (large κ_r) may result from a negative ϵ_1 , as in a plasma below its plasma frequency, and not necessarily from absorption (large ϵ_2). In other words, energy may be reflected rather than absorbed.

One might guess that $\epsilon_1(\omega)$ and $\epsilon_2(\omega)$ are independent functions, since one describes polarization and the other dissipation, which are apparently distinct physical processes. This turns out to be quite wrong: $\epsilon_2(\omega)$ is entirely determined if $\epsilon_1(\omega)$ is known for all frequencies and vice versa. This remarkable result is known as the Kramers–Kronig relation and is described in Appendix 6. We shall use it in Section 10.1 to derive the imaginary part of $\tilde{\sigma}$ from its real part, which is easier to calculate.

8.5.2 ELECTRON-PHOTON INTERACTION

The strategy of the calculation is to calculate the energy absorbed from a plane wave of light in two ways, one using Fermi's golden rule and the other from classical electromagnetism; equating the two gives the real part of the conductivity σ_1 . Apply a light wave with an electric field $\mathbf{E}(\mathbf{R}, t) = 2eE_0 \cos(\mathbf{Q} \cdot \mathbf{R} - \omega t)$. Assume that the wave does not decay as it passes through the sample. There is a magnetic field too but its effect is usually much smaller and we shall ignore it. The unit vector $\mathbf{e}$ gives the polarization of the electric field, which must lie in the plane normal to the direction of propagation, so $\mathbf{e} \perp \mathbf{Q}$. This wave is plane polarized but a complex $\mathbf{e}$ can be used to describe more general waves. Classically, the electric field induces a current whose in-phase component is $\mathbf{J} = \sigma_1 \mathbf{E}$, which dissipates energy at a rate $\mathbf{J} \cdot \mathbf{E}$ per unit volume. Averaging over time, the total power dissipated in volume Ω is

$$P = 2\sigma_1 \Omega E_0^2. \quad (8.65)$$

We shall now derive a quantum-mechanical expression using Fermi's golden rule.

The Hamiltonian contains scalar and vector potentials rather than the fields directly (Section 6.1). Here we use $\mathbf{E} = -\partial\mathbf{A}/\partial t$, which requires a vector potential $\mathbf{A}(\mathbf{R}, t) = (2eE_0/\omega) \sin(\mathbf{Q} \cdot \mathbf{R} - \omega t)$. A scalar potential would give the wrong kind of field. Consider a single Fourier component, $\phi(\mathbf{R}, t) \propto \exp[i(\mathbf{Q} \cdot \mathbf{R} - \omega t)]$. This gives $\mathbf{E} = -\text{grad } \phi \propto i\mathbf{Q} \exp[i(\mathbf{Q} \cdot \mathbf{R} - \omega t)]$. The important feature is that the electric field is in the same direction as the direction of travel $\mathbf{Q}$. This is called a longitudinal field in contrast to the transverse nature of an electromagnetic wave.

The Schrödinger equation for an electron (charge $-e$) is

$$\left[\frac{(\hat{\mathbf{p}} + e\mathbf{A})^2}{2m_0} + V_{\text{crystal}} \right] \Psi = i\hbar \frac{\partial \Psi}{\partial t}, \quad (8.66)$$

where V_{crystal} is the potential of the crystal or heterostructure under study. The perturbation due to the light wave is the difference between this Hamiltonian and that without $\mathbf{A}$, which gives

$$\hat{V} = \frac{e}{2m_0} (\mathbf{A} \cdot \hat{\mathbf{p}} + \hat{\mathbf{p}} \cdot \mathbf{A} + e\mathbf{A}^2). \quad (8.67)$$

Note that the order of $\mathbf{A}$ and $\hat{\mathbf{p}}$ has been respected, as $\hat{\mathbf{p}}$ is an operator and cannot be moved around like a simple number. The term with $\mathbf{A}^2$ is of second order in the electric field and will be neglected; it can be shown rigorously to vanish within the electric-dipole approximation that we shall shortly make. The $\hat{\mathbf{p}} \cdot \mathbf{A}$ term also needs a little thought, remembering that it is a differential operator that acts on the wave function Ψ . It means

$$\hat{\mathbf{p}} \cdot \mathbf{A}\Psi = -i\hbar \nabla \cdot (\mathbf{A}\Psi) = -i\hbar [(\nabla \cdot \mathbf{A})\Psi + \mathbf{A} \cdot (\nabla \Psi)], \quad (8.68)$$

where the rule for the divergence of the product of a vector and a scalar has been used. Now $\nabla \cdot \mathbf{A} = 0$ for our transverse wave so only $\mathbf{A} \cdot \hat{\mathbf{p}}\Psi$ remains and the total perturbation is $\hat{V} = (e/m_0)\mathbf{A} \cdot \hat{\mathbf{p}}$. This form of interaction may look rather odd, but remember that the current operator (1.42) is something like $(e/m_0)\hat{\mathbf{p}}$. Thus our perturbation can be written roughly in the form $\mathbf{A} \cdot \mathbf{J}$, which is used in classical electrodynamics and is closely related to the simplest expression $\mathbf{E} \cdot \mathbf{J}$.

Expanding the perturbation gives

$$\hat{V}(\mathbf{R}, t) = \frac{eE_0}{im_0\omega} [e^{i(\mathbf{Q} \cdot \mathbf{R} - \omega t)} - e^{-i(\mathbf{Q} \cdot \mathbf{R} - \omega t)}](\mathbf{e} \cdot \hat{\mathbf{p}}). \quad (8.69)$$

This is very close to the form (8.47) of the electron-phonon perturbation. The first term represents the absorption of a photon by the electron, increasing the electron's energy by $\hbar\omega$ and its momentum by $\hbar\mathbf{Q}$. Similarly, the second term describes the emission of a photon so the electron loses both energy and momentum.

In optical phenomena the formulas can usually be simplified because the photon carries negligible momentum and induces vertical transitions in bands (Section 2.7).

Typical optical experiments are done with light of visible or nearby wavelengths, which have wavelengths in the hundreds of nanometres. This is much longer than a unit cell in a semiconductor crystal or the width of a quantum well, which set the scale of the electronic wave functions. Thus we can usually neglect the momentum of the photon and set $\mathbf{Q} = \mathbf{0}$ in the perturbation, treating the electric field as constant across the electronic states. This is the *electric-dipole approximation* (and is the opposite of the quasi-elastic approximation that we made for acoustic phonons).

We can now put the perturbation into Fermi's golden rule. The transition rate from a state i to another state j due to the absorption of a photon is

$$W_{ji} = \frac{2\pi}{\hbar} \left(\frac{eE_0}{m_0\omega} \right)^2 |\langle j | \mathbf{e} \cdot \hat{\mathbf{p}} | i \rangle|^2 \delta(E_j - E_i - \hbar\omega). \quad (8.70)$$

The matrix element looks a little complicated. The middle expands to

$$\mathbf{e} \cdot \hat{\mathbf{p}} = -i\hbar \left(e_x \frac{\partial}{\partial x} + e_y \frac{\partial}{\partial y} + e_z \frac{\partial}{\partial z} \right). \quad (8.71)$$

For example, if the electric field were polarized purely along z , $\mathbf{e} = (0, 0, 1)$ and $\mathbf{e} \cdot \hat{\mathbf{p}} = -i\hbar \partial/\partial z$.

The power absorbed by this transition is the product of the transition rate and the energy of each photon $\hbar\omega$. The next step is to sum over all possible initial and final states to get the total power. We must include Fermi factors $f(E_i)$ to ensure that the initial state is filled and $[1 - f(E_j)]$ to ensure that the final state is empty, otherwise the transition cannot take place. Thus the system absorbs power at a rate

$$P_+ = \frac{2\pi}{\hbar} \hbar\omega \left(\frac{eE_0}{m_0\omega} \right)^2 2 \sum_{i,j} |\langle j | \mathbf{e} \cdot \hat{\mathbf{p}} | i \rangle|^2 f(E_i) [1 - f(E_j)] \delta(E_j - E_i - \hbar\omega). \quad (8.72)$$

The factor of 2 in front of the summation is for spin, which must be counted only once as the spin is not changed by the optical transition (this factor is always a nuisance and it is difficult to know the best place for it; conventions vary).

The system also emits energy at a rate P_- due to the $e^{+i\omega t}$ term in the perturbation. This is identical to P_+ except for the change in sign of ω :

$$P_- = -\frac{2\pi}{\hbar} \hbar\omega \left(\frac{eE_0}{m_0\omega} \right)^2 2 \sum_{i,j} |\langle j | \mathbf{e} \cdot \hat{\mathbf{p}} | i \rangle|^2 f(E_i) [1 - f(E_j)] \delta(E_j - E_i + \hbar\omega). \quad (8.73)$$

The argument of the δ -function can be restored to the form in P_+ by interchanging the labels i and j , which are only dummy summation indices. The matrix elements are not affected but the Fermi functions change to $f(E_j)[1 - f(E_i)]$. These factors and the overall sign are now the only differences between P_+ and P_- , and together they give

$$f(E_i)[1 - f(E_j)] - f(E_j)[1 - f(E_i)] = f(E_i) - f(E_j). \quad (8.74)$$

Thus the overall rate of absorption of energy is

$$P = \frac{2\pi}{\hbar} \hbar \omega \left(\frac{e E_0}{m_0 \omega} \right)^2 2 \sum_{i,j} |\langle j | \mathbf{e} \cdot \hat{\mathbf{p}} | i \rangle|^2 [f(E_i) - f(E_j)] \delta(E_j - E_i - \hbar \omega). \quad (8.75)$$

Comparison with the classical result (8.65) shows that

$$\sigma_1(\omega) = \frac{\pi e^2}{m_0^2 \omega} \frac{2}{\Omega} \sum_{i,j} |\langle j | \mathbf{e} \cdot \hat{\mathbf{p}} | i \rangle|^2 [f(E_i) - f(E_j)] \delta(E_j - E_i - \hbar \omega). \quad (8.76)$$

This is the general formula for the real part of the conductivity, which can be converted to the imaginary part of the dielectric function using equation (8.64). The factor of 2 is written in front of the summation as a reminder that it arises from the sum over spins. Although the volume Ω of the system appears in the denominator, it must cancel after the summation as it did for impurity scattering. The mass in equation (8.76) is m_0 , that for *free* electrons, rather than an effective mass. This is because it comes from the kinetic energy term of the original Schrödinger equation rather than an effective mass approximation. We shall derive the corresponding formula for ϵ_1 or σ_2 in Section 10.1.

We argued earlier that we normally expect $\sigma_1 > 0$. The only factors that affect the sign are ω and the occupation functions. The δ -function forces $E_j > E_i$ (if $\omega > 0$) and a system in equilibrium has $f(E_i) > f(E_j)$. This guarantees that $\sigma_1 > 0$ and the system will absorb energy from light. It is easy to show that σ_1 is an even function of ω and is therefore always positive in equilibrium. In a laser, on the other hand, one arranges for the population of some states to be *inverted* so that $f(E_j) > f(E_i)$ and in this case the system can release energy to amplify an incident beam of light (Section 10.6).

8.6 Interband Absorption

We saw in Section 2.7 that optical absorption is a convenient way of measuring direct band gaps and can now calculate this for the simple conduction and valence bands shown in Figure 8.4. Assume that the material is three-dimensional, undoped, and at zero temperature so the valence band is full and the conduction band is empty. The sums over states i and j in equation (8.76) now each become a sum over the bands with another over the Bloch wave vector $\mathbf{K}$ within each band. In our example we can assume that the initial state is in the valence band, $i = v\mathbf{K}$, and the final state is in the conduction band, $j = c\mathbf{K}'$, as the occupation factors give 1 in this case but vanish otherwise. Thus

$$\sigma_1(\omega) = \frac{\pi e^2}{m_0^2 \omega} \frac{2}{\Omega} \sum_{\mathbf{K}, \mathbf{K}'} |\langle c\mathbf{K}' | \mathbf{e} \cdot \hat{\mathbf{p}} | v\mathbf{K} \rangle|^2 \delta[\epsilon_c(\mathbf{K}') - \epsilon_v(\mathbf{K}) - \hbar \omega]. \quad (8.77)$$

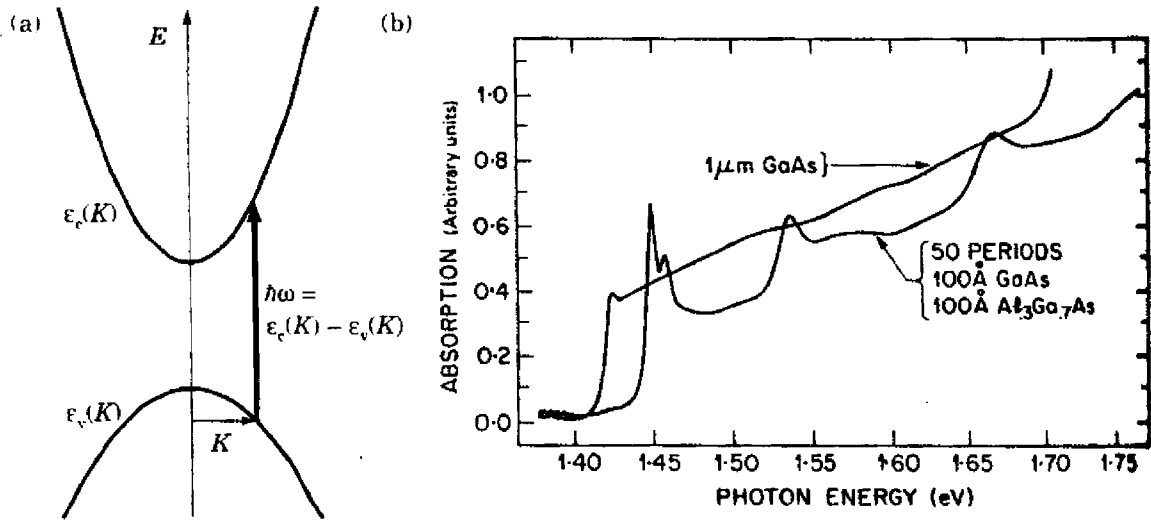


FIGURE 8.4. (a) Absorption from the valence band to the conduction band in a semiconductor with a direct band gap. (b) Absorption of two- and three-dimensional samples of GaAs at room temperature. The active volume of the two-dimensional (multiquantum well) sample is only half that of the three-dimensional one. [Reprinted from Schmitt-Rink, Chemla, and Miller (1989) by kind permission of Taylor and Francis.]

Look at the matrix element first. The Bloch states can be written in the form

$$\psi_{v\mathbf{K}}(\mathbf{R}) = \Omega^{-1/2} e^{i\mathbf{K} \cdot \mathbf{R}} u_{v\mathbf{K}}(\mathbf{R}), \quad (8.78)$$

where $u_{v\mathbf{K}}(\mathbf{R})$ is periodic from one unit cell of the crystal to the next. The factor of $\Omega^{-1/2}$ normalizes the states in such a way that $u_{v\mathbf{K}}(\mathbf{R}) = 1$ in the limit of free electrons. The operator $\hat{\mathbf{p}} = -i\hbar\nabla$ takes the first derivative, so the product rule gives

$$\hat{\mathbf{p}}\psi_{v\mathbf{K}}(\mathbf{R}) = \Omega^{-1/2} e^{i\mathbf{K} \cdot \mathbf{R}} [\hbar\mathbf{K}u_{v\mathbf{K}}(\mathbf{R}) + \hat{\mathbf{p}}u_{v\mathbf{K}}(\mathbf{R})]. \quad (8.79)$$

Thus the matrix element is

$$\langle c\mathbf{K}' | \mathbf{e} \cdot \hat{\mathbf{p}} | v\mathbf{K} \rangle = \frac{1}{\Omega} \mathbf{e} \cdot \int u_{c\mathbf{K}'}^*(\mathbf{R}) e^{i(\mathbf{K}-\mathbf{K}') \cdot \mathbf{R}} [\hbar\mathbf{K}u_{v\mathbf{K}}(\mathbf{R}) + \hat{\mathbf{p}}u_{v\mathbf{K}}(\mathbf{R})] d^3\mathbf{R}. \quad (8.80)$$

The first term in square brackets vanishes after integration because it reduces to the product of two states that must be orthogonal. The remaining term also vanishes unless $\mathbf{K}' = \mathbf{K}$ because of the plane waves in the wave functions; the proof is along the same lines as the discussion of umklapp scattering earlier. We now have $\langle c\mathbf{K}' | \mathbf{e} \cdot \hat{\mathbf{p}} | v\mathbf{K} \rangle = \delta_{\mathbf{K}', \mathbf{K}} p_{cv}(\mathbf{K})$, where

$$p_{cv}(\mathbf{K}) = \frac{1}{\Omega} \int u_{c\mathbf{K}}^*(\mathbf{R}) (\mathbf{e} \cdot \hat{\mathbf{p}}) u_{v\mathbf{K}}(\mathbf{R}) d^3\mathbf{R}. \quad (8.81)$$

We can again use the fact that $u(\mathbf{R})$ is periodic between unit cells to reduce the integral to the volume of a unit cell rather than the crystal,

$$p_{cv}(\mathbf{K}) = \frac{1}{\Omega_{\text{cell}}} \int_{\text{cell}} u_{c\mathbf{K}}^*(\mathbf{R}) (\mathbf{e} \cdot \hat{\mathbf{p}}) u_{v\mathbf{K}}(\mathbf{R}) d^3\mathbf{R}. \quad (8.82)$$

The expression for the conductivity now becomes

$$\sigma_1(\omega) = \frac{\pi e^2}{m_0^2 \omega} \frac{2}{\Omega} \sum_{\mathbf{K}} |p_{cv}(\mathbf{K})|^2 \delta[\varepsilon_c(\mathbf{K}) - \varepsilon_v(\mathbf{K}) - \hbar\omega]. \quad (8.83)$$

This is the general result for σ_1 in a crystal except that we have removed the occupation functions and sums over bands, which can easily be restored.

We have several times assumed that the periodic part of the Bloch functions are only weakly dependent on $\mathbf{K}$ near the edge of a band (equation 3.5), and put $u_{n\mathbf{K}} \approx u_{n0}$. The matrix element reduces to $p_{cv}(0)$ if we repeat this assumption here, and can be pulled out of the summation. It leaves

$$\sigma_1(\omega) \approx \frac{\pi e^2}{m_0^2 \omega} |p_{cv}(0)|^2 \left\{ \frac{2}{\Omega} \sum_{\mathbf{K}} \delta[\varepsilon_c(\mathbf{K}) - \varepsilon_v(\mathbf{K}) - \hbar\omega] \right\}. \quad (8.84)$$

The summation in braces defines a density of states per unit volume (Section 1.7.3) known as the *optical joint density of states* $n_{\text{opt}}(E)$. It involves the energies of two bands but is almost identical in form to the usual result and contains a single factor of 2 for spin. The conductivity can now be written in the simple form

$$\sigma_1(\omega) \approx \frac{\pi e^2}{m_0^2 \omega} |p_{cv}(0)|^2 n_{\text{opt}}(\hbar\omega). \quad (8.85)$$

The matrix element is the same as that in $\mathbf{k} \cdot \mathbf{p}$ theory (Section 7.3) and we can write $p_{cv}(0) = (im_0/\hbar)P$. The strength of absorption is closely related to other quantities that depend on P such as the effective masses, and σ_1 can be rewritten in terms of E_P , which varies little among the common semiconductors. The differences arise from the density of states.

The optical joint density of states looks much like an ordinary density of states, and its behaviour dominates $\sigma_1(\omega)$. Suppose that the two bands are parabolic and spherically symmetric in the region of interest, an acceptable approximation near Γ in GaAs, with

$$\varepsilon_v(\mathbf{K}) = E_v - \frac{\hbar^2 K^2}{2m_0 m_h}, \quad \varepsilon_c(\mathbf{K}) = E_c + \frac{\hbar^2 K^2}{2m_0 m_e}. \quad (8.86)$$

Then the difference of energies that appears in the definition of $n_{\text{opt}}(E)$ is

$$\varepsilon_c(\mathbf{K}) - \varepsilon_v(\mathbf{K}) = (E_c - E_v) + \frac{\hbar^2 K^2}{2m_0} \left(\frac{1}{m_e} + \frac{1}{m_h} \right). \quad (8.87)$$

Therefore $n_{\text{opt}}(E)$ will be the familiar density of states for free electrons but with the bottom of the band offset to the band gap $E_g = E_c - E_v$ and with the optical joint effective mass given by the reduced mass $1/m_{\text{eh}} = 1/m_e + 1/m_h$. In three dimensions the density of states rises from zero like $(E - E_g)^{1/2}$, and equation (8.85)

shows that the optical conductivity just above the absorption edge at $\hbar\omega = E_g$ should do the same.

Data for GaAs are shown in Figure 8.4(b). Unfortunately, the shape of the absorption curve does not agree well with our predictions. The problem is caused by excitons again and will be addressed in Section 10.7.

This calculation can be extended to low-dimensional systems if we don't worry too much about the matrix element and polarization of the light (Chapter 10). The major change is in $n_{\text{opt}}(E)$, which behaves like the density of states for free electrons (Figure 1.9). Thus the absorption edge in a quantum well should be a steplike feature, and a wire should show a divergence like $(\hbar\omega - E_g)^{-1/2}$. Again these results are not quantitatively correct but the trend is extremely important: the optical absorption edge becomes a stronger feature as the number of dimensions is reduced. The absorption edge is also shifted to higher energies to include the energy of the electron and hole above the bottom of the quantum well. The energy levels are discrete in a quantum dot and the absorption reflects this, being a series of lines rather than a band.

8.7 Absorption in a Quantum Well

As a contrasting example, consider absorption between bound states in the quantum well aligned along z , shown in Figure 8.5. This system is immersed in the conduction band of a heterostructure and we ought to use effective-mass wave functions. We shall defer this complication to Section 10.5 as it has no qualitative effect on the behaviour.

We know from Section 4.5 that the states factorize into a product of a bound state in z and a transverse plane wave,

$$\psi_{ik}(\mathbf{R}) = A^{-1/2} \phi_i(z) \exp(i\mathbf{k} \cdot \mathbf{r}), \quad E_i(\mathbf{k}) = \varepsilon_i + \frac{\hbar^2 k^2}{2m}, \quad (8.88)$$

where $\mathbf{r} = (x, y)$ and A is the area of the quantum well. Let L be the thickness of the whole sample (not just the well) along z so the volume $\Omega = AL$. Each state i for motion along z gives rise to a subband of energies.

Consider the matrix element between two such states $\langle j\mathbf{k}' | \mathbf{e} \cdot \hat{\mathbf{p}} | i\mathbf{k} \rangle$. The form of this depends strongly on $\mathbf{e}$, the polarization of the light. Suppose first that $\mathbf{e} = (1, 0, 0)$ so that the electric field is parallel to x , in the plane of the quantum well. The direction of propagation might be along y or z , in the plane of the quantum well or normal to it (see Figure 10.6 for examples of polarization and propagation). In this case, $\mathbf{e} \cdot \hat{\mathbf{p}} = -i\hbar \partial/\partial x$ and

$$(\mathbf{e} \cdot \hat{\mathbf{p}}) \psi_{ik}(\mathbf{R}) = \hbar k_x \psi_{ik}(\mathbf{R}) \quad (8.89)$$

because the momentum operator affects only the plane wave in $\psi_{ik}(\mathbf{R})$. Then

$$\langle j\mathbf{k}' | \mathbf{e} \cdot \hat{\mathbf{p}} | i\mathbf{k} \rangle = \hbar k_x \langle j\mathbf{k}' | i\mathbf{k} \rangle = 0; \quad (8.90)$$



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